INCH-POUND
MIL-PRF-3098/114C
May 17, 2004
SUPERSEDING
MIL-PRF-3098/114B
1 October 1997

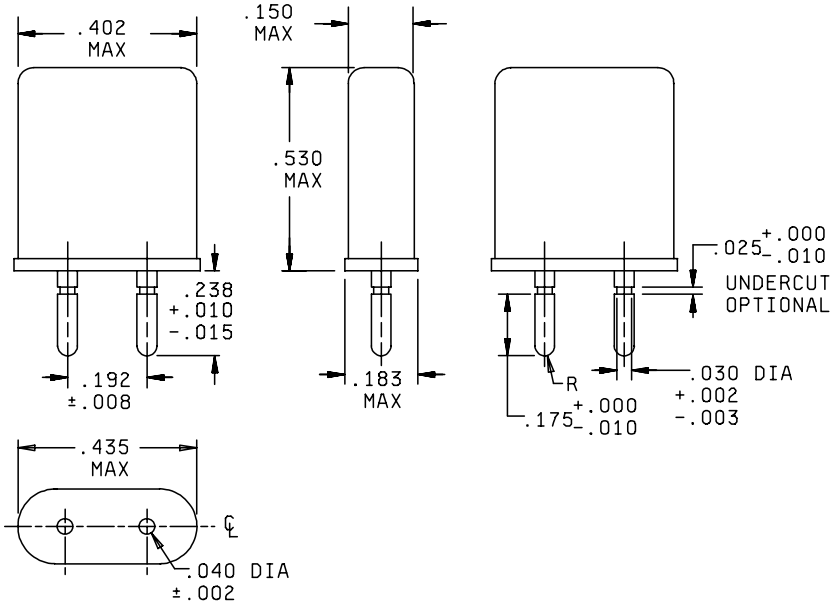
PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR135/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-3098.

Pertinent characteristics: 10 MHz to 12 MHz; fundamental; controlled; antiresonance.



Inches	mm	Inches	mm	Inches	mm	Inches	mm
.002	.05	.015	.38	.150	3.81	.238	6.05
.003	.08	.025	.64	.175	4.45	.402	10.21
.008	.20	.030	.76	.183	4.65	.435	11.05
.010	.25	.040	1.02	.192	4.88	.530	13.46

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR135/U.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 10 MHz to 12 MHz, inclusive.

Capacitance, shunt: 7.0 pF, maximum.

Mode of oscillation: Fundamental.

Rated drive level: 1.0 mW, maximum.

Calibration values:

Equivalent resistance: 11 ohms.

Crystal current: 15 mA.

Antiresonance, load capacitance: 32.0 pF ±0.5 pF.

Reference temperature: +85°C ±2°C .

Operating temperature range: +80°C to +90°C, inclusive.

Frequency tolerance: ±0.002 percent.

Equivalent resistance: 22 ohms, maximum.

Frequency stability: ±0.0001 percent.

Operable temperature range: -55°C to +80°C, inclusive.

Shock:

Frequency change permitted: 4 Hz, maximum.

Equivalent resistance: ±10 percent.

Vibration: Method 204, MIL-STD-202, test condition A.

Frequency change permitted: 4 Hz, maximum.

Equivalent resistance change permitted: ±10 percent.

Temperature cycling:

Frequency: Normal frequency ±0.002 percent.

Equivalent resistance: 22 ohms, maximum.

Aging:

Frequency change permitted: ±0.0005 percent.

Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Air Force - 19, 99

Preparing activity:
DLA - CC

(Project 5955-0758)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at www.dodssp.daps.mil.

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